

Aidan Patrick Reddy

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EDUCATION

Massachusetts Institute of Technology	Cambridge, MA
Ph.D. in Physics	9/2021 - 5/2026
Columbia College, Columbia University	New York, NY
B.A. in Physics with a Concentration in Mathematics, <i>magna cum laude</i>	9/2018 - 4/2021
Swarthmore College	Swarthmore, PA
transferred	9/2017 - 5/2018

HONORS & FELLOWSHIPS

GLAM Postdoctoral Fellowship, Stanford University	7/2026 - Present
Jane Street Graduate Research Fellowship Finalist	5/2025
Kavli Institute for Theoretical Physics Graduate Fellowship	7/2024 - 12/2024
Phi Beta Kappa	4/2021
Columbia College Physics Department Honors	4/2021
NSF GRFP Honorable Mention	2021 & 2022
Columbia College Work Exemption Program Grant	fall 2019 & spring 2020
Dean's List	all semesters

RESEARCH EXPERIENCE

GLAM Postdoctoral Fellow, Stanford University	Stanford, CA
Advisor: Trithep Devakul	7/2026 - Present
Graduate Research Assistant, MIT Department of Physics	Cambridge, MA
Advisor: Liang Fu	4/2022 - 5/2026
Project title: <i>Electron fractionalization in moiré superlattices</i>	
Research Experience for Undergraduates, Columbia University MRSEC	New York, NY
Advisor: Allan H. MacDonald	6/2020 - 9/2021
Project title: <i>Resonant Coulomb energy transfer in transition metal dichalcogenide moirés</i>	
Undergraduate Research Assistant, Columbia Department of Physics	New York, NY
Advisor: Cory Dean	1/2019 - 3/2020
Project title: <i>Achieving precise twist-angle control and homogeneity in twisted bilayer graphene devices</i>	
Science Undergraduate Laboratory Internship, SLAC National Lab	Menlo Park, CA
Advisors: Ryan Davis, Apurva Mehta	6/2019 - 8/2019
Project title: <i>Correction of self-absorption distortion in X-ray absorption near-edge spectroscopy</i>	

TEACHING EXPERIENCE

Teaching Assistant, MIT Department of Physics	Cambridge, MA
Classical Mechanics	fall 2025
Quantum Physics II	spring 2025
Theory of Solids I	fall 2022 & fall 2023
Junior Lab II	spring 2022
Teaching Assistant, Swarthmore College Department of Mathematics	Swarthmore, PA
Single-Variable Calculus II	spring 2018

PUBLICATIONS

*co-first author

1. **A. P. Reddy** and L. Fu, "Quantum melting a Wigner crystal into Hall liquids." [Physical Review B 113.16 \(2026\)](#).
2. **A. P. Reddy**, D. N. Sheng, A. Abouelkomsan, E. J. Bergholtz, and L. Fu, "Anti-topological crystal and non-Abelian liquid in twisted semiconductor bilayers", [Nature Communications, in press. \(2026\)](#).
3. B. A. Foutty, **A. P. Reddy**, C. R. Kometter, K. Watanabe, T. Taniguchi, T. Devakul, and B. E. Feldman, "Magnetic Hofstadter cascade in a twisted semiconductor homobilayer", [Nat. Phys., 1–7 \(2025\)](#).
4. E. Anderson, J. Cai, **A. P. Reddy**, H. Park, W. Holtzmann, K. Davis, T. Taniguchi, K. Watanabe, T. Smolenski, A. Imamoğlu, T. Cao, D. Xiao, L. Fu, W. Yao, and X. Xu, "Trion sensing of a zero-field composite Fermi liquid", [Nature 635, 590–595 \(2024\)](#).
5. H. Li, Z. Xiang, **A. P. Reddy**, T. Devakul, R. Sailus, R. Banerjee, T. Taniguchi, K. Watanabe, S. Tongay, A. Zettl, L. Fu, M. F. Crommie, and F. Wang, "Wigner molecular crystals from multielectron moiré artificial atoms", [Science 385, 86–91 \(2024\)](#).
6. B. A. Foutty, C. R. Kometter, T. Devakul, **A. P. Reddy**, K. Watanabe, T. Taniguchi, L. Fu, and B. E. Feldman, "Mapping twist-tuned multiband topology in bilayer WSe₂", [Science 384, 343–347 \(2024\)](#).
7. T. Tan*, **A. P. Reddy***, L. Fu, and T. Devakul, "Designing topology and fractionalization in narrow gap semiconductor films via electrostatic engineering", [Phys. Rev. Lett. 133, 206601 \(2024\)](#).
8. **A. P. Reddy***, N. Paul*, A. Abouelkomsan, and L. Fu, "Non-Abelian fractionalization in topological minibands", [Phys. Rev. Lett. 133, 166503 \(2024\)](#).
[Featured in Physics](#)

9. D. N. Sheng, **A. P. Reddy**, A. Abouelkomsan, E. J. Bergholtz, and L. Fu, “Quantum anomalous Hall crystal at fractional filling of moiré superlattices”, *Phys. Rev. Lett.* **133**, 066601 (2024).

Editors’ Suggestion

10. A. Abouelkomsan, **A. P. Reddy**, L. Fu, and E. J. Bergholtz, “Band mixing in the quantum anomalous Hall regime of twisted semiconductor bilayers”, *Phys. Rev. B* **109**, L121107 (2024).
11. Z. Lu, T. Han, Y. Yao, **A. P. Reddy**, J. Yang, J. Seo, K. Watanabe, T. Taniguchi, L. Fu, and L. Ju, “Fractional quantum anomalous Hall effect in multilayer graphene”, *Nature* **626**, 759–764 (2024).
12. **A. P. Reddy**, T. Devakul, and L. Fu, “Artificial atoms, Wigner molecules, and an emergent kagome lattice in semiconductor moiré superlattices”, *Phys. Rev. Lett.* **131**, 246501 (2023).

Editors’ Suggestion

13. **A. P. Reddy** and L. Fu, “Toward a global phase diagram of the fractional quantum anomalous Hall effect”, *Phys. Rev. B* **108**, 245159 (2023).

Editors’ Suggestion

14. H. Goldman*, **A. P. Reddy***, N. Paul*, and L. Fu, “Zero-field composite Fermi liquid in twisted semiconductor bilayers”, *Phys. Rev. Lett.* **131**, 136501 (2023).

Featured in Physics, Editors’ Suggestion

15. **A. P. Reddy**, F. Alsallom, Y. Zhang, T. Devakul, and L. Fu, “Fractional quantum anomalous Hall states in twisted bilayer MoTe₂ and WSe₂”, *Phys. Rev. B* **108**, 085117 (2023).

Editors’ Suggestion

16. C. R. Kometter, J. Yu, T. Devakul, **A. P. Reddy**, Y. Zhang, B. A. Foutty, K. Watanabe, T. Taniguchi, L. Fu, and B. E. Feldman, “Hofstadter states and re-entrant charge order in a semiconductor moiré lattice”, *Nat. Phys.* **19**, 1861–1867 (2023).

PREPRINTS

1. N. Paul*, A. Abouelkomsan*, **A. P. Reddy***, and L. Fu, “Shining light on collective modes in moiré fractional Chern insulators”, [arXiv:2502.17569](https://arxiv.org/abs/2502.17569) (2025).
2. D. Luo, **A. P. Reddy**, T. Devakul, and L. Fu, “Artificial intelligence for artificial materials: moiré atom”, [arXiv:2303.08162](https://arxiv.org/abs/2303.08162) (2023).

TALKS

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|---|---------|
| 1. American Physical Society Global Physics Summit
<i>Quantum melting a Wigner crystal into Hall liquids</i> | 3/2026 |
| 2. Flatiron Institute Center for Computational Quantum Physics Seminar (invited)
<i>Quantum melting a Wigner crystal into Hall liquids</i> | 10/2025 |
| 3. Stanford Condensed Matter Physics Seminar (invited)
<i>Quantum melting a Wigner crystal into Hall liquids</i> | 10/2025 |
| 4. UCLA Quantum Seminar (invited)
<i>Fractional quantum anomalous Hall effects in twisted MoTe₂ bilayers</i> | 3/2025 |
| 5. American Physical Society Global Physics Summit 2025 (invited symposium)
<i>Fractional quantum anomalous Hall effects in twisted MoTe₂ bilayers</i> | 3/2025 |
| 6. Harvard Condensed Matter Theory Kids’ Seminar (invited)
<i>Shining light on collective modes in fractional quantum anomalous Hall states</i> | 3/2025 |
| 7. Cornell University Condensed Matter Theory Seminar (invited)
<i>Non-Abelian fractional Chern insulators and competing states in twisted MoTe₂ bilayers</i> | 1/2025 |
| 8. Università di Pisa Condensed Matter Physics Seminar (invited)
<i>Fractional quantum anomalous Hall effects in twisted semiconductor bilayers</i> | 11/2024 |
| 9. Kavli Institute for Theoretical Physics Moiré Workshop Seminar (invited)
<i>Non-Abelian fractional Chern insulator in twisted semiconductor bilayers</i> | 9/2024 |
| 10. Kavli Institute for Theoretical Physics Condensed Matter Theory Seminar (invited)
<i>Topology and fractionalization in moiré materials</i> | 8/2024 |
| 11. Stanford GLAM Seminar (invited)
<i>Quantum anomalous Hall regime in twisted semiconductor bilayers</i> | 5/2024 |
| 12. American Physical Society March Meeting
<i>Toward a global phase diagram of the fractional quantum anomalous Hall effect</i> | 4/2024 |
| 13. Physical Review Journal Club (invited)
<i>Zero-field composite Fermi liquid in twisted semiconductor bilayers</i> | 11/2023 |
| 14. MIT Condensed Matter Theory Seminar (invited)
<i>Fractional quantum anomalous Hall states in semiconductor moiré homobilayers</i> | 5/2023 |
| 15. American Physical Society March Meeting
<i>Moiré resonant energy transfer</i> | 3/2022 |
| 16. Extraordinary Electronic Switching of Thermal Transport MURI Collaboration (invited)
<i>Resonant Energy Transfer in TMD Moirés</i> | 8/2021 |
| 17. Columbia Undergraduate Science Journal, Columbia University (invited) | 11/2020 |

<i>X-Ray Absorption Spectroscopy “Self-Absorption” Correction</i>		
18. Cory Dean Lab Meeting, Columbia University		10/2020
<i>Journal Club on Mapping Local Heterogeneity in Open-Faced Twisted Bilayer Graphene Devices</i>		
19. Arun Majumdar Lab Meeting, Stanford University (invited)		8/2020
<i>Energy transfer via Coulomb Scattering in twisted bilayer Transition Metal Dichalcogenides</i>		
20. Cory Dean Lab Meeting, Columbia University (invited)		8/2020
<i>Energy transfer via Coulomb Scattering in twisted bilayer Transition Metal Dichalcogenides</i>		
21. MRSEC REU Presentation, Columbia University		7/2020
<i>Energy transfer via Coulomb Scattering in twisted bilayer Transition Metal Dichalcogenides</i>		
22. Cory Dean Lab Meeting, Columbia University		4/2020
<i>Nematicity and Competing Orders in Superconducting Magic-Angle Graphene</i>		
23. Solid State Physics Course, Columbia University		12/2019
<i>A Stack, a Twist, and a Hint of “Magic”: Correlated Physics in twisted bilayer Graphene</i>		
24. Society of Physics Students, Columbia University		10/2019
<i>A Stack, a Twist, and a Hint of “Magic”: Correlated Physics in twisted bilayer Graphene</i>		
25. SULI Program Final Presentation, SLAC National Accelerator Laboratory		8/2019
<i>X-Ray Absorption Spectroscopy “Self-Absorption” Correction</i>		
26. Cory Dean Lab Meeting, Columbia University		4/2019
<i>Optimizing the Homogeneity of Twisted Bilayer Graphene Devices</i>		

POSTERS

1. Northeast Quantum Forum 2025: AI in Quantum	10/2025
<i>Quantum melting a Wigner crystal into Hall liquids</i>	
2. University of Colorado at Boulder Summer School for Condensed Matter Physics	7/2025
<i>Wigner crystals and integer quantum Hall states in the two-dimensional electron gas</i>	
3. Thouless Institute Winter Workshop, University of Washington	1/2024
<i>Toward a global phase diagram of the fractional quantum anomalous Hall effect</i>	
4. Quantum Geometry in Condensed Matter Workshop (Beverly, MA)	10/2023
<i>Fractional quantum anomalous Hall regime in twisted semiconductor bilayers</i>	
5. Dynamical Response and Transport in Quantum Magnets workshop, KITP	8/2023
<i>Fractional quantum anomalous Hall regime in twisted semiconductor bilayers</i>	
6. Quantum materials group meeting, Canadian Institute for Advanced Research	5/2023
<i>Fractional quantum anomalous Hall states in semiconductor moiré homobilayers</i>	
7. Topology, symmetry, and interactions in crystals workshop, KITP	4/2023
<i>Moiré atoms, Wigner molecules, and emergent Kagome lattice</i>	
8. Frontiers of Quantum Materials and Devices Conference (Valencia, Spain)	6/2022
<i>Electron-assisted hopping in semiconductor moirés</i>	

MEDIA COVERAGE

- Thomson, E. A. (2024, November 18). [MIT physicists predict exotic form of matter with potential for quantum computing](#). MIT News.
- Wilkinson, R. (2024, October 17). [Quantum Computing with a Twist](#). Physics Magazine.
- Chu, J. (2024, February 21). [Electrons become fractions of themselves in graphene, study finds](#). MIT News.
- Fadelli, I. (2024, September 2). [Exploring new physics arising from electron interactions in semiconductor moiré structures](#). Phys.org.
- Fadelli, I. (2024, September 1). [Study predicts a new quantum anomalous crystal in fractionally filled moiré superlattices](#). Phys.org.
- Duque, T. (2024, November 7). [Wigner molecular crystals from multielectron moiré artificial atoms](#). Berkeley Lab News Center.
- Hadhazy, A. (2024, May 8). [A “magic” angle between layers in a stacked nanoscale system offers intriguing material properties](#). Stanford School of Humanities and Sciences News.
- Jain, J. (2023, September 27). [In a twist, composite fermions form and flow without a magnetic field](#). APS Physics Magazine.
- Feldman, B.E. (2023, September 18). [Competing electron solids and electron fluids in the moiré atomic limit](#). Nature Physics.